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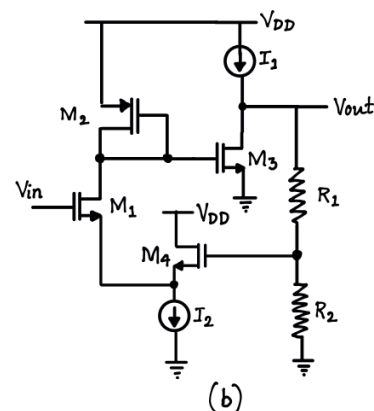
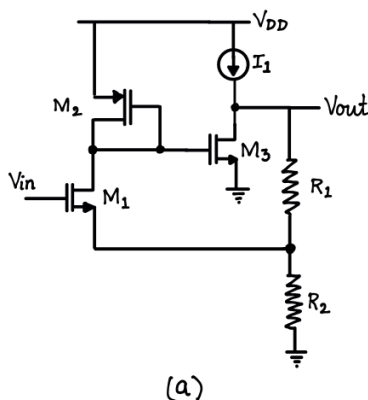
Recheck Request :**Instructions :**

- Please answer only in the space provided. Anything written outside will not be evaluated.
- Clearly show all steps used to arrive at final answer. Final answer is to be written in the space provided for it.
- Please note that answers without proper units and sign will be marked wrong.
- Overwriting will not be evaluated. Please keep your work legible, clean and crisp.

Marks obtained:

Question No.	Marks
1	/10
2	/24
3	/14
4	/7
5	/5
Total	/60

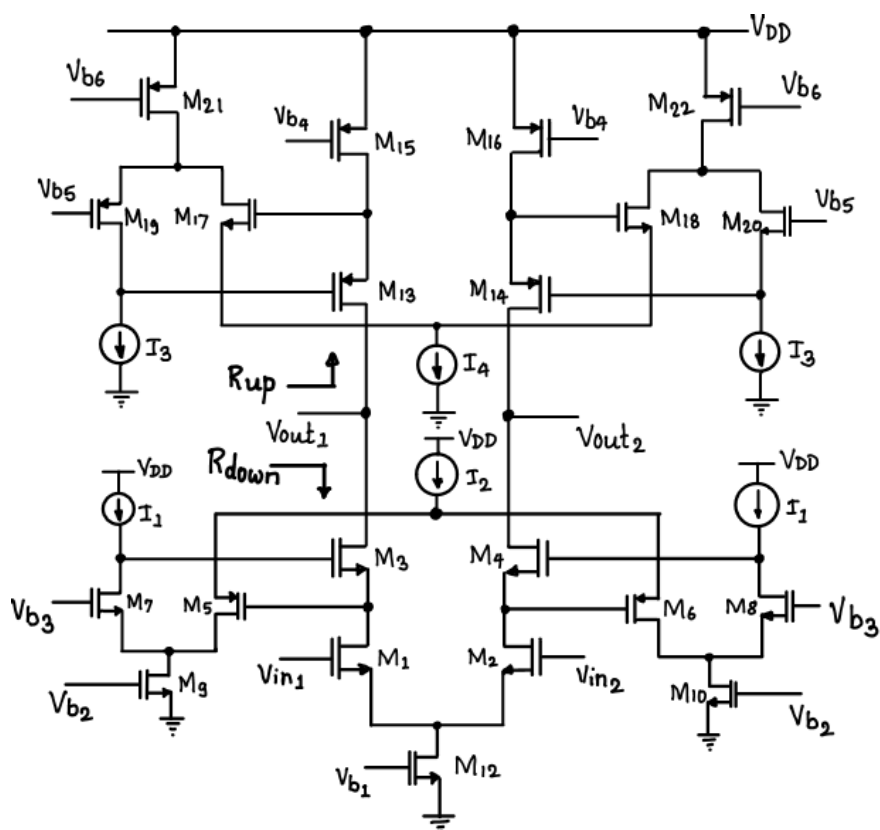
Q1. Design the circuits shown below for an open-loop voltage gain of 20, an open-loop output impedance of 4 k Ω , and a closed-loop voltage gain of 4. Assume $\lambda=0$. $g_{m1}=g_{m3}=g_{m4}=8g_{m2}$. (Find out values of g_{m1} , g_{m2} , R_1 , R_2 and Feedback factor). Assume current sources I_1 , I_2 are ideal.



Important Steps :

	Figure (a)	Figure(b)
Feedback Factor	0.2	0.2
g_{m1}	1.04 mS	1.248 mS
g_{m2}	0.13 mS	0.156 mS
R_1	3.2 k Ω	3.2 k Ω
R_2	800 Ω	800 Ω

Q2. For the circuit given below, fill the table below. For all MOSFETS, $g_m = 1$ mS, $r_o = 10$ k Ω , $|V_{OV}| = 0.1$ V, $|V_{th}| = 0.3$ V. Minimum voltage required across current sources I_1 and I_3 is $2V_{OV}$ and across current sources I_2 and I_4 is V_{OV} . All current sources are ideal except minimum voltage requirement across them. Neglect body effect. $V_{DD} = 2$ V.



Important Steps :

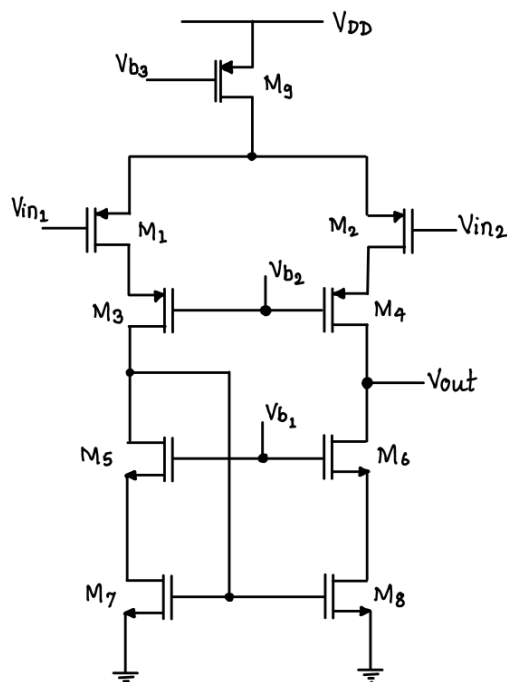
V_{b4}	$V_{b5,max}$	V_{b6}	V_{b1}	V_{b2}	$V_{b3,min}$
1.6 V	1.5 V	1.6 V	0.4 V	0.4 V	0.5 V
G_m of circuit	R_{up}	R_{down}	R_{out} of circuit	Gain	Maximum output voltage swing
1 mS	5 M Ω	5 M Ω	2.5 M Ω	2500 V/V	3 V

Q3. Design high voltage swing telescopic cascode amplifier shown in figure with following specifications

- All PMOS are of approximately same size and all NMOS are of same size.
- Slew rate is 5 V/ μs for load capacitor of 10 pF.
- $V_{out,max} = 1.28$ V and $V_{out,min} = 0.3$ V for $V_{DD} = 2$ V.

Technology parameters are : $\mu_n C_{ox} = 100 \mu A/V^2$, $\mu_p C_{ox} = 50 \mu A/V^2$, $V_{thn} = 0.3$ V, $V_{thp} = -0.4$ V, $\lambda_n = 0.1$ V $^{-1}$, and $\lambda_p = 0.2$ V $^{-1}$ for channel length of 1 μm .

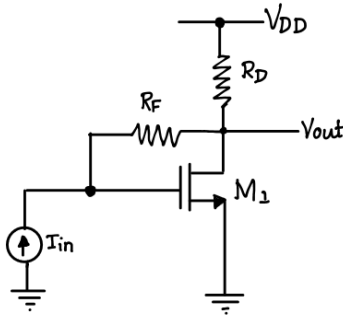
Calculate the sizes of all transistors, $V_{b1,min}$, $V_{b2,max}$ and V_{b3} . Neglect body effect. What is the gain of circuit?



Important Steps :

$(W/L)_{1-4}$	$(W/L)_{5-8}$	$(W/L)_9$	$V_{b1,min}$	$V_{b2,max}$	V_{b3}	Voltage Gain
22.67	22 22	22.22	0.6 V	0.88 V	1.3 V	1921

Q4. For the circuit shown below, $R_D=50\text{ k}\Omega$, $R_F=20\text{ k}\Omega$, $g_{m1}=1\text{ mS}$ and $r_{o1}=100\text{ k}\Omega$. Fill the table given below. Do not use miller approximation method.



Important Steps :

Feedback factor	$-0.05\text{ mS} = \frac{-1}{20\text{K}}$
Open loop input impedance	$20\text{K}\Omega$
Open loop output impedance	$12.50\text{K}\Omega$
Open loop gain	$-250\text{K}\Omega$
Closed loop input impedance	$1.481\text{K}\Omega$
Closed loop output impedance	926Ω
Closed loop gain	$-18.516\text{K}\Omega$

Q5.

- (a) If the phase margin of the amplifier is 50° , then the magnitude of closed loop gain of amplifier used in feedback with feedback factor β at gain crossover frequency (ω_{GX}) is given by

$$\left| \frac{V_{out}}{V_{in}} (j\omega_{GX}) \right| = \frac{K}{\beta}$$

What is the value of K ?

- (b) In Miller compensated OTA, for phase margin of 45° , what is the relation required between load capacitor C_L and compensation capacitor C_c ; if location of right half plane Zero is at $(10 \times \text{Gain Bandwidth Product})$?

Important Steps :

Q5(a)	Q5(b)
Value of K	Relation between C_L and C_c
1.183	$C_c \geq 0.122 C_L$